

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Systems for Data Science, Optimization in CS, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, Java, C++, SQL, Bash, HTML/CSS

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, async services, integration testing, authentication, rate limiting, multi-tenant SaaS

AI & GenAI

LLM agents, OpenAI / Groq APIs, RAG, retrieval & provenance, PyTorch, HuggingFace Transformers, Claude Code / Copilot / Cursor

Databases & Web

MongoDB, PostgreSQL, Redis, SQLite, React, Electron, GraphQL

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered a **multi-tenant NHS clinical eTriage platform** (patient self-referral and triage) for **8+ UK clinical clients** (Node.js, MongoDB), processing **10,000+ monthly submissions** as production healthcare infrastructure integrated with Mayden's iaptus platform via REST APIs, contributing to **\$340K+** in revenue.
- Architected a full-stack annotation platform (**React, Node.js, MongoDB**) used daily by the AI team, replacing a slow manual spreadsheet workflow with secure role-based REST APIs and cutting **100+ hours/month** of manual work while improving throughput by **60%**.
- Owned backend reliability across the eTriage stack: monitored production releases and client integrations, shipped fixes for tenant-specific incidents, hardened authentication and rate limiting, and remediated **20+ VAPT findings** to protect sensitive clinical data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** and prototyped **LLM-powered context-aware replies** in a chat product, improving maintainability and end-user response quality.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **AI agent** for web discovery that plans typed retrieval, reranks pages, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns **cell-level provenance** so every value is traceable to its source.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, and pipeline instrumentation, keeping agent behavior predictable and easy to debug.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **cross-platform desktop app** integrating camera, microphone, UI, and backend into an on-device pipeline (MediaPipe hand tracking, whisper.cpp ASR, local TTS), designed and shipped end-to-end under hackathon time pressure.
- Engineered a **FastAPI backend** and **Electron/React frontend** that run fully offline; designed async APIs and queues to keep the UI responsive while handling local video and audio streams.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, WebSockets, SQL

- Built an **edge-native AI agent** that fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time, prototyped end-to-end on Cloudflare Workers.
- Designed durable workflow boundaries so multi-step agent tasks survive process restarts and tool failures; agent state and conversation history persist server-side without an external database.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.